THE LINNIK PROBLEM FOR A SINGLE IDEAL CLASS: EQUIDISTRIBUTION OF THE CLASSES AND THE VARIANCE OF THE CLASS DEVIATION

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Abstract. For a prime discriminant $-D$ with class number $h > 1$ and a class C of binary forms Q_C of discriminant $-D$ we consider $S_C(N) = \sum_{p<N} r_C(N-p)$, the number of representations $N = p + Q_C(x,y)$. The class decomposition $r_C = r_{\mathrm{gen}} + \frac{w}{h}\sum_{\psi\neq 1}\bar\psi(C)c_\psi$ of Part V splits S_C into the genus part, which is the Linnik problem for $\frac{w}{h}(1*\chi_{-D})$ and has, under the hypothesis H(N) of the first note, the main term $\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N)$ *independently of h and of the class*, and the class deviations $S_\psi(N) = \sum_{p<N} c_\psi(N-p)$, sums of the coefficients of the weight-one dihedral cusp forms ϑ_ψ over the shifted primes. We prove unconditionally that $\sum_{N\le X}|S_\psi(N)|^2 \ll X^2\log X$, so that for all $N \le X$ outside $O(X^{1-2\eta}\log X)$ exceptions the counts of the different classes differ by at most $N^{1/2+\eta}$, and the classes are asymptotically equidistributed for all $N \le X$ but $O((\log X)^4)$; under H(N) and the dual Titchmarsh bound of Assing–Blomer–Li for ϑ_ψ every class has the asymptotic formula for every large N. Numerically ($D = 23, 31$, $N \le 4\cdot 10^6$) both classes agree with the genus formula to 10^{-4} and $S_\psi(N) \approx 0.12\sqrt{N}$. The second moment is not the diagonal: $\sum_{p<N} c_\psi(N-p)^2$ is $\kappa_\psi\pi(N)$ on average, with the Rankin–Selberg residue κ_ψ of Part V, while the mean square of $S_\psi(N)$ is 0.41–0.43 of it and rising. We explain this: by modularity the exponential sum of ϑ_ψ is exponentially small at the rationals a/q with $q^2 \ll X$ (Lemma 3.1), and these major arcs carry half of the L^2 mass of the prime exponential sum; we conjecture that the mean square is $\sim \frac{1}{2}\kappa_\psi\pi(N)$.

1. Setting and results

Let $D \equiv 3 \pmod 4$ be a prime, $D > 3$, $K = \mathbb{Q}(\sqrt{-D})$, $\chi = \chi_{-D}$, $w = 2$, $h = h(-D)$ (odd), and for a class $C \in \mathrm{Cl}(K)$ let $r_C(n) = \#\{(x,y) \in \mathbb{Z}^2 : Q_C(x,y) = n\}$ as in Part V [3]. With $c_\psi(n) = \sum_{N\mathfrak{a}=n}\psi(\mathfrak{a})$ ($\psi \in \widehat{\mathrm{Cl}}$, real, $c_{\bar\psi} = c_\psi$) and $r_{\mathrm{gen}} = \frac{w}{h}(1*\chi)$,

$$r_C = r_{\mathrm{gen}} + \frac{w}{h}\sum_{\psi\neq 1}\bar\psi(C)\,c_\psi, \qquad S_C(N) := \sum_{p<N} r_C(N-p) = S_{\mathrm{gen}}(N) + \frac{w}{h}\sum_{\psi\neq 1}\bar\psi(C)\,S_\psi(N), \tag{1}$$

where $S_{\mathrm{gen}}(N) = \sum_{p<N} r_{\mathrm{gen}}(N-p)$ and $S_\psi(N) = \sum_{p<N} c_\psi(N-p)$; $\vartheta_\psi = \sum c_\psi(n)e(nz) \in S_1(\Gamma_0(D),\chi)$ is the dihedral cusp form of ψ (Part V, Theorem 1.1). For $h = 3$, $S_{C_0} = S_{\mathrm{gen}} + \frac{4}{3}S_c$, $S_{C_1} = S_{\mathrm{gen}} - \frac{2}{3}S_c$ with $c = c_\psi$, and $S_{C_0} - S_{C_1} = 2S_c$: for $D = 23$, $S_c(N)$ is half the difference between the numbers of representations $N = p + x^2 + xy + 6y^2$ and $N = p + 2x^2 + xy + 3y^2$.

1.1. **The genus part.** Since $r_{\mathrm{gen}} = \frac{w}{h}(1*\chi)$ and the first note [2] treats $r_D = w_D(1*\chi)$ for $h = 1$, its proof applies verbatim with w_D replaced by $\frac{w}{h}$, and the constant becomes $\frac{w}{h}L(1,\chi) = \frac{2\pi}{\sqrt{D}}$ by the class number formula:

Proposition 1.1. *Under the hypothesis* H(N) *of* [2, Hypothesis 1.1] *(the level $\sqrt{N}(\log N)^A$ of Bombieri–Friedlander–Iwaniec for the single residue class N),*

$$S_{\mathrm{gen}}(N) = \frac{2\pi}{\sqrt{D}}\,\mathfrak{S}_D(N)\,\pi(N) + O\Big(N\Big(\frac{\log\log N}{\log N}\Big)^2\Big),$$

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$$\mathfrak{S}_D(N)=\Big(1-\frac{\chi(N)}{D-1}\Big)\prod_{\ell\mid N}\Big(1-\frac{\chi(\ell)}{\ell}\Big)\prod_{\ell\nmid N}\Big(1+\frac{\chi(\ell)}{\ell(\ell-1)}\Big).$$

The main term is the same as for the principal form of a field of class number one with the same D: the class number cancels between the genus average and the class number formula, exactly as the leading coefficient of the scalloped norms in Part V. Every class has the local densities of the genus (there is one genus), so the singular series is the same for all classes, and (1) reduces the Linnik problem for a single class to the size of $S_\psi(N)$.

1.2. **The class deviation.**

Lemma 1.2 (Wilton-type bound)**.** *Let $f=\sum a(n)e(nz)\in S_1(\Gamma_0(D),\chi)$. Then, uniformly in $\alpha\in\mathbb{R}$ and $Y\geq 2$, $\sum_{n\leq Y}a(n)e(n\alpha)\ll_f\sqrt{Y}\log Y$, and $\sum_n a(n)e^{-n/Y}e(n\alpha)=f(\alpha+\frac{i}{2\pi Y})\ll_f\sqrt{Y}$.*

Proof. The function $y^{1/2}|f(z)|$ is $\Gamma_0(D)$-invariant ($|\chi|=1$) and tends to 0 at both cusps, so it is bounded on $\mathbb{H}$: $|f(x+iy)|\leq C_f y^{-1/2}$. This is the second statement. For the first, with $y=1/Y$, $\sum_{n\leq Y}a(n)e(n\alpha)=\int_0^1 f(\alpha+\beta+iy)K(\beta)\,\mathrm{d}\beta$, $K(\beta)=\sum_{m\leq Y}e^{2\pi m/Y}e(-m\beta)$; by partial summation against the Dirichlet kernels, whose L^1 norms are $\ll\log Y$, and since $e^{2\pi m/Y}$ is increasing and $\leq e^{2\pi}$, $\|K\|_1\ll\log Y$. □

Theorem 1.3. *For every $\psi\neq 1$ and $X\geq 2$,*

$$\sum_{N\leq X}|S_\psi(N)|^2\ll_D X^2\log X.$$

Consequently, for every $\eta>0$, all $N\leq X$ outside a set of $O_D(X^{1-2\eta}\log X)$ elements satisfy $|S_C(N)-S_{\mathrm{gen}}(N)|\leq N^{1/2+\eta}$ for every class C; and for every $\varepsilon>0$, all $N\leq X$ outside a set of $O_{D,\varepsilon}((\log X)^4)$ elements satisfy $|S_C(N)-S_{\mathrm{gen}}(N)|\leq\varepsilon N/\log N$ for every class.

Proof. For $N\leq X$, $S_\psi(N)$ is the coefficient of $e(N\alpha)$ in $P(\alpha)F(\alpha)$ with $P(\alpha)=\sum_{p\leq X}e(p\alpha)$ and $F(\alpha)=\sum_{1\leq m\leq X}c_\psi(m)e(m\alpha)$, since $p<N$ and $m=N-p$ are both $\leq X$. By Parseval and Lemma 1.2,

$$\sum_{N\leq X}|S_\psi(N)|^2\leq\int_0^1|P(\alpha)F(\alpha)|^2\,\mathrm{d}\alpha\leq\|F\|_\infty^2\int_0^1|P|^2=\|F\|_\infty^2\,\pi(X)\ll X(\log X)^2\frac{X}{\log X}.$$

By (1) the same bound holds for S_C-S_{gen}. The number of $N\in(X/2,X]$ with $|S_C(N)-S_{\mathrm{gen}}(N)|>(X/2)^{1/2+\eta}$ is $\ll X^2\log X/X^{1+2\eta}$, and with $>\varepsilon X/(2\log X)$ it is $\ll(\log X)^3/\varepsilon^2$; summing over dyadic ranges gives the two counts. □

Thus the equidistribution of the representations among the classes holds unconditionally for almost all N, with a power saving, while the asymptotic formula itself, which is a statement about the genus part, needs $\mathrm{H}(N)$ already for $h=1$. For individual N the class deviation needs an input of a different kind, a bound for the coefficients of a cusp form along $N-p$; this is the dual problem of Titchmarsh type.

Hypothesis 1.4 (T_ψ)**.** *$S_\psi(N)=o(N/\log N)$ as $N\to\infty$.*

Assing, Blomer and Li [1, Theorem 1.3] prove, for a holomorphic or Maaß cuspidal newform for $\Gamma_0(M)$ with Hecke eigenvalues $\lambda(n)$, $\sum_{Y<n\leq X}\Lambda(n)\lambda(f-n)\ll X^{1-\delta}$ for $f>X$, $X+f\asymp X$ (the case $\sigma=-1$ of their theorem, which they describe as the cuspidal analogue of the dual sum $\sum_{p<N}\tau(N-p)$ and which was out of reach before their work). With $f=N$, $Y=0$, $X=N-1$ their theorem bounds $\sum_{n<N}\Lambda(n)\lambda(N-n)$ by $N^{1-\delta}$, and partial summation gives T_ψ with a power saving, provided (i) the theorem extends to the weight-one forms ϑ_ψ, which have the nebentypus χ_{-D} (their statement is for $\Gamma_0(M)$ with trivial character), and (ii) it holds for the truncations

$n \le t$, $N/(\log N)^A \le t \le N$, i.e. with $X + f \ll X(\log X)^A$ instead of $X + f \asymp X$. Remark 3.5 records what (i) requires; (ii) we have not checked. We therefore state:

Theorem 1.5. *Under* $\mathrm{H}(N)$ *and* T_ψ *for all* $\psi \neq 1$*, for every class* C*,*

$$S_C(N) = \frac{2\pi}{\sqrt{D}}\,\mathfrak{S}_D(N)\,\pi(N)\,(1+o(1)),$$

the same main term for every class; in particular every large N *is* $p + Q_C(x,y)$ *for every class* C*, and the representations are equidistributed among the classes.*

Proof. Proposition 1.1, (1), T_ψ, and $\mathfrak{S}_D(N) \gg 1$ [2, Theorem 1.2]. □

2. Numerics

Table 1 compares S_{C_0}, S_{C_1} with the main term of Proposition 1.1 for $D = 23$ and 31, on 400 random N in each range (script `linnik_class.py`, exact counts by FFT from the representation numbers). The class deviation is of square-root size, $\operatorname{rms} S_c(N) \approx 0.12\sqrt{N}$ (a fit of $\log \operatorname{rms}$ against $\log N$ gives the exponent 0.46 for both D), and it has no bias: its mean over $N \in [2\cdot 10^6, 4\cdot 10^6)$ is $0.000\sqrt{N}$ separately for $\chi(N) = 1, -1, 0$.

	$D = 23$				$D = 31$			
N	S_{C_0}/main	S_{C_1}/main	$\frac{\operatorname{rms} S_c}{\sqrt{N}}$	$\frac{E\,S_c^2}{\kappa_D \pi(N)}$	S_{C_0}/main	S_{C_1}/main	$\frac{\operatorname{rms} S_c}{\sqrt{N}}$	$\frac{E\,S_c^2}{\kappa_D \pi(N)}$
$[10^5, 2\cdot 10^5)$	1.0001	1.0001	0.126	0.405	0.9997	0.9998	0.131	0.414
$[3\cdot 10^5, 6\cdot 10^5)$	1.0002	0.9999	0.125	0.414	1.0000	1.0003	0.126	0.422
$[10^6, 2\cdot 10^6)$	1.0005	1.0003	0.119	0.423	1.0001	1.0000	0.119	0.430
$[2\cdot 10^6, 4\cdot 10^6)$	0.9999	1.0000	0.113	0.427	1.0002	0.9999	0.119	0.432

Table 1. The Linnik problem for the two classes of $D = 23$ ($x^2 + xy + 6y^2$, $2x^2 + xy + 3y^2$) and $D = 31$ ($x^2 + xy + 8y^2$, $2x^2 + xy + 4y^2$): the counts against $\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N)$ (means over 400 random N; standard deviations 0.002–0.03), the class deviation $S_c = \frac{1}{2}(S_{C_0} - S_{C_1})$, and its mean square over the whole range against $\kappa_D\pi(N)$, $\kappa_{23} = 0.421799$, $\kappa_{31} = 0.430026$ (`linnik_class_variance.py`). The diagonal $\sum_{p<N} c(N-p)^2$ is $\kappa_D\pi(N)$ to 10^{-4} in every range; for $D = 23$ and $N \in [8\cdot 10^6, 1.6\cdot 10^7)$ the last column is 0.419 for $\chi(N) = 1$, 0.448 for $\chi(N) = -1$, 0.433 for $\chi(N) = 0$.

3. The variance of the class deviation

The diagonal of $S_\psi(N)^2$ is $\sum_{p<N} c_\psi(N-p)^2$, and since $\sum_{m\le x} c_\psi(m)^2 = \kappa_\psi x + O(x^{4/5})$ with the residue κ_ψ of Part V, Theorem 1.3(iii), its average over $N \le X$ is $\kappa_\psi\pi(N)$ (Table 1: to 10^{-4}). The mean square of $S_\psi(N)$ is much smaller, 0.41–$0.43\,\kappa_\psi\pi(N)$, increasing slowly with N. The off-diagonal correlations $\sum_{p\neq p'} c_\psi(N-p)c_\psi(N-p')$ are therefore negative on average and of the size of the diagonal. The mechanism is visible on the circle.

Lemma 3.1 (Depletion at the major arcs)**.** *Let* $f \in S_1(\Gamma_0(D), \chi)$*,* $F_X(\alpha) = \sum_n a(n)e^{-n/X}e(n\alpha) = f(\alpha + \frac{i}{2\pi X})$*, and* $(a,q) = 1$*. Then*

$$|F_X(a/q)| \ll_f \frac{X}{q}\exp\Big(-\frac{4\pi^2 X}{Dq^2}\Big) \quad (D \nmid q), \qquad |F_X(a/q)| \ll_f \frac{X}{q}\exp\Big(-\frac{4\pi^2 X}{q^2}\Big) \quad (D \mid q).$$

Proof. Let $\gamma = \left(\begin{smallmatrix} a & b \\ q & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $z = a/q + iy = \gamma w$, so that $w = -d/q + i/(q^2 y)$ and $|qw + d| = 1/(qy)$. Then $f(z) = (qw + d)\,(f|_1\gamma)(w)$. If $D \mid q$, $\gamma \in \Gamma_0(D)$ and $f|_1\gamma = \chi(d)f$; if $D \nmid q$, γ maps ∞ to a cusp equivalent to 0, and $f|_1\gamma$ is the expansion of f at 0, $\sum_{n\geq 1} b(n)e(nw/D)$ ($f|W_D = -if$ for the dihedral forms, Part V, Theorem 1.1). In both cases $|f|_1\gamma(w)| \ll e^{-2\pi \operatorname{Im} w/D'}$ with $D' = D$ or 1, and $\operatorname{Im} w = 1/(q^2 y) = 2\pi X/q^2$. □

So F_X vanishes to exponential precision at every a/q with $q^2 \leq \delta X/\log X$, whereas its mean square over $[0,1]$ is $\sum c_\psi(n)^2 e^{-2n/X} \sim \frac{1}{2}\kappa_\psi X$. Numerically, with $\rho(q)$ the mean of $|F_X(a/q)|^2$ over $(a,q) = 1$ divided by this mean square ($D = 23$, $X = 3\cdot 10^5$, `linnik_class_arcs.py`): $\rho(q) < 10^{-4}$ for $q^2 < 0.26X$, then 0.008, 0.04, 0.12, 0.50 at $q^2/X = 0.45$, 0.59, 0.77, 1.34, an overshoot to 1.50 at $q^2/X \approx 7$, and $\rho \to 1$ (e.g. 1.07 at $q^2/X = 36$).

The prime exponential sum does the opposite: $|P(a/q+\beta)|^2 \approx \frac{\mu(q)^2}{\varphi(q)^2}|T(\beta)|^2$ with $\int |T|^2 \approx X/(\log X)^2$, so the arcs at the $\varphi(q)$ fractions a/q carry the L^2 mass $\frac{\mu(q)^2}{\varphi(q)}\frac{X}{(\log X)^2}$, and the arcs with $q \leq Q$ carry $\frac{X}{(\log X)^2}\sum_{q\leq Q}\frac{\mu(q)^2}{\varphi(q)} = \frac{X}{(\log X)^2}(\log Q + 1.3325 + o(1))$, out of the total $\pi(X) \approx X/\log X$. (The mean square of the primes in the residue classes modulo q, summed over $q \leq Q$, is asymptotically known for $Q \leq X$ by the theorem of Barban–Davenport–Halberstam in the form of Montgomery [4] and Hooley [5]; it is the quantity that measures how much of the mass of $|P|^2$ the major arcs carry.) The arcs with $q \leq \sqrt{X}$ carry half the mass, and on them F is exponentially small. Writing $\int |P|^2|F|^2$ as the diagonal minus the deficit on the major arcs gives the prediction

$$\frac{E\,|S_\psi(N)|^2}{\kappa_\psi \pi(N)} \approx 1 - \frac{\sum_q \mu(q)^2\varphi(q)^{-1}(1-\rho(q))}{\log X + 1.3325} = \frac{\frac{1}{2}\log X + O(1)}{\log X + 1.3325},$$

which for $X = 3\cdot 10^5$ evaluates to $1 - 7.449/13.94 = 0.466$, against the measured 0.41–0.42 in the ranges around $3\cdot 10^5$. The model is only qualitative at these sizes. For the smoothed variance $V(X) = \sum_N e^{-2N/X} S_\psi(N)^2$, which equals $\int_0^1 |P|^2|F|^2$ exactly for $P = \sum_p e^{-p/X}e(p\alpha)$ and $F = F_X$, the exact ratio to the diagonal is 0.393 and 0.403 at $X = 3\cdot 10^4$ and 10^5 (`linnik_class_smooth.py`), while the same model with $|F(a/q+\beta)|^2$ integrated against the width $|T(\beta)|^2$ of the arcs gives 0.499 and 0.498: the integration over β weakens the depletion ($\rho = 0.21$ instead of $< 10^{-4}$ at $q^2/X = 0.1$, because $\operatorname{Im} w$ in Lemma 3.1 drops by the factor $1 + 4\pi^2X^2\beta^2$ across the arc), and the model then overestimates V by 0.1, an $O(1/\log X)$ error it does not capture (the normalisation of the arc masses with the weights $1/\log n$, and the local structure at D: the measured ratio is 0.412 for $\chi(N) = 1$ and 0.442 for $\chi(N) = -1$ around $3\cdot 10^5$, the diagonal itself being 0.955 and 1.045 times $\kappa_D\pi(N)$ because $m = N - p$ avoids the class N modulo D). The leading constant $\frac{1}{2}$ does not depend on these details: the transition region and the overshoot change only the $O(1)$, and fitting $V(\log N + 1.3325) = a\log N + b$ to the four ranges of Table 1 gives $a = 0.52$ for $D = 23$, and the constant $b = V(\log N + 1.3325) - \frac{1}{2}\log N$ is -0.59, -0.57, -0.53, -0.52, -0.52 at $N \approx 1.5\cdot 10^5$, $4.5\cdot 10^5$, $1.5\cdot 10^6$, $3\cdot 10^6$, $1.2\cdot 10^7$: stable, as the limit $\frac{1}{2}$ requires (a different limit a would make b drift linearly in $\log N$); the two smoothed values give the slope $a = 0.50$ as well.

Conjecture 3.2. *For every $\psi \neq 1$, as $X \to \infty$,*

$$\sum_{X<N\leq 2X} |S_\psi(N)|^2 \sim \frac{1}{2}\,\kappa_\psi \sum_{X<N\leq 2X} \pi(N),$$

and the same with the exponent $\frac{1}{2}$ for the coefficients of any fixed holomorphic cusp form (with κ_ψ replaced by its Rankin–Selberg residue, normalised to weight one): the class deviation of the Linnik problem has half the variance of a sum of independent terms.

Remark 3.3. (a) The half is a property of the circle, not of the class group: it comes from the modularity of f, which makes F small at the rationals of height up to $\sqrt{X}$, and from the logarithmic distribution of the mass of $|P|^2$ over the denominators. Section 3.1 reduces the conjecture to two statements about the primes alone and proves everything else. (b) For $h = 1$ the analogue of S_ψ does not exist (there is no cusp form), and for composite D with several genera the genus characters give Eisenstein series, for which F is large at the major arcs and the class deviation has a main term (Part V, Section 7): the class equidistribution of Theorem 1.3 is a phenomenon of the single genus. (c) For $h = 3$, Theorem 1.3 and Conjecture 3.2 say that $\#\{N = p+x^2+xy+6y^2\} - \#\{N = p+2x^2+xy+3y^2\} = 2S_c(N)$ is typically of size $\sqrt{2\kappa_{23}N/\log N} = 0.92\sqrt{N/\log N}$, against a main term of size $1.31\,N/\log N$ for each.

3.1. **What is proved about the variance.** We work with the smoothed objects $P(\alpha) = \sum_p e^{-p/X}e(p\alpha)$, $F(\alpha) = F_X(\alpha) = \sum_m c_\psi(m)e^{-m/X}e(m\alpha)$ and

$$V(X) := \sum_N e^{-2N/X} S_\psi(N)^2 = \int_0^1 |P|^2|F|^2,$$

$$\mathrm{Diag}(X) := \sum_N e^{-2N/X} \sum_{p<N} c_\psi(N-p)^2 = \int_0^1 |P|^2 \int_0^1 |F|^2,$$

both identities exact ($e^{-p/X}e^{-m/X} = e^{-N/X}$; the second is Parseval for each factor), so V/Diag is the correlation coefficient of $|P|^2$ and $|F|^2$ on the circle, and $\mathrm{Diag} \sim \frac{\kappa_\psi}{2} X \cdot \frac{X}{2\log X}$. For $B \geq 3$ let

$$Q = X^{1/2}(\log X)^{-B}, \qquad K = (\log X)^{(B-1)/2},$$

$$\mathfrak{M} = \bigcup_{q\leq Q} \bigcup_{(a,q)=1} \Big[\frac{a}{q} - \frac{K}{X}, \frac{a}{q} + \frac{K}{X}\Big], \qquad \mathfrak{m} = [0,1) \setminus \mathfrak{M}$$

(the arcs are disjoint, $2K/X < Q^{-2}$).

Theorem 3.4. $\sup_{\alpha\in\mathfrak{M}} |F(\alpha)| \ll_A X^{-A}$ *for every* A*; hence* $V(X) = \int_{\mathfrak{m}} |P|^2|F|^2 + O_A(X^{-A})$.

Proof. Let $\alpha = \frac{a}{q} + \beta \in \mathfrak{M}$, $z = \alpha + iy$, $y = \frac{1}{2\pi X}$, so that $F(\alpha) = f(z)$, and $\gamma = \left(\begin{smallmatrix} a & b \\ q & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, $z = \gamma w$. As in Lemma 3.1, $|f(z)| = |f|_1\gamma(w)|/(q|\beta + iy|) \leq |f|_1\gamma(w)|/(qy)$ with

$$\mathrm{Im}\, w = \frac{y}{q^2(\beta^2+y^2)} \geq \frac{2\pi X}{q^2(1+4\pi^2K^2)} \geq \frac{2\pi(\log X)^{2B}}{1+4\pi^2(\log X)^{B-1}} \gg (\log X)^{B+1},$$

and $|f|_1\gamma(w)| \ll e^{-2\pi\,\mathrm{Im}\, w/D}$ for $\mathrm{Im}\, w \geq 1$ (the expansion at the cusp $\gamma\infty$ has width 1 or D and polynomially bounded coefficients). So $|F(\alpha)| \ll X\exp(-c(\log X)^{B+1})$, and $\int_{\mathfrak{M}} |P|^2|F|^2 \leq \sup_{\mathfrak{M}} |F|^2 \int_0^1 |P|^2$. □

Thus the major arcs of order $X^{1/2-\varepsilon}$ contribute nothing to the variance, whatever the primes do there. Conjecture 3.2 (in the smoothed form $V \sim \frac{1}{2}\mathrm{Diag}$) follows from the two statements

(M) $$\int_{\mathfrak{M}} |P|^2 = \Big(\frac{1}{2} + o(1)\Big) \int_0^1 |P|^2 \qquad (\text{for some fixed } B, \text{ then } o(1) = O(\log\log X/\log X)),$$

(D) $$\int_{\mathfrak{m}} |P|^2|F|^2 = (1+o(1)) \int_{\mathfrak{m}} |P|^2 \cdot \int_0^1 |F|^2,$$

and conversely (M) and the conjecture imply (D). Statement (M) is the Hardy–Littlewood prediction for the mass of the arcs: with $P^*(\frac{a}{q}+\beta) = \frac{\mu(q)}{\varphi(q)}T(\beta)$, $T(\beta) = \sum_{n\geq 2} \frac{e^{-n/X}}{\log n} e(n\beta)$, one has $\int_{\mathfrak{M}} |P^*|^2 = \sum_{q\leq Q} \frac{\mu(q)^2}{\varphi(q)} \int_{|\beta|\leq K/X} |T|^2 = (\log Q + O(1)) \frac{X}{2(\log X)^2}(1 + O(K^{-1}))$, which is ($\frac{1}{2}$ –

$\frac{B\log\log X}{\log X}+O(\frac{1}{\log X}))\int_0^1|P|^2$. A proof of (M) needs $\int_{\mathfrak{M}}|P-P^*|^2=o(X/\log X)$ for arcs of order $X^{1/2-\varepsilon}$. By Gallagher's lemma and the expansion of $e(an/q)$ in characters this is a mean value $\sum_{r\le Q}\frac{r}{\varphi(r)}\log\frac{Q}{r}\sum^*_{\chi \bmod r}|\psi(t+H,\chi)-\psi(t,\chi)|^2\ll H^2(\log X)^{-A}$ over intervals of length $H=X/2K$: a Barban–Davenport–Halberstam estimate in short intervals with the weight r instead of $1/\varphi(r)$. The large sieve gives $(H+Q^2)H\log X$, the trivial size, and the Bombieri–Vinogradov theorem, Hooley's form of the Barban–Davenport–Halberstam theorem and the mean-square Hardy–Littlewood theorem of Lavrik and Mikawa for prime pairs all lose a power of Q against this weight; under GRH the pointwise bound $P-P^*\ll(qX)^{1/2}(\log X)^2(1+X|\beta|)$ gives (M) for arcs of order $X^{1/3-\varepsilon}$ only, i.e. the mass $\frac{1}{3}$. So (M) is open, unconditionally and under GRH. Statement (D) is a genuine correlation statement between the primes and the cusp form on the minor arcs, of the kind that the pair correlation hypothesis supplies in the theorem of Goldston and Montgomery [6] on the variance of the primes in short intervals, $\int_1^X(\psi(x+\delta x)-\psi(x)-\delta x)^2\,\mathrm{d}x\sim\frac{1}{2}\delta X^2\log\frac{1}{\delta}$: there the deficit $\log\frac{1}{\delta}$ against $\log X$ is the mass of the arcs of order δ^{-1} on which the short interval kernel vanishes, as here it is the mass of the arcs of order $\sqrt{X}$ on which F vanishes, and the analogue of $\delta=X^{-1/2}$ gives the $\frac{1}{2}$.

Numerically (`linnik_class_split.py`, $D=23$, the arcs of order $c\sqrt{X}$ with $K=1$; the c below are fixed, so the arcs include part of the transition zone of ρ):

X	V/Diag	$\int_{\mathfrak{M}}\lvert P\rvert^2/\int\lvert P\rvert^2$, $c=\frac{1}{4}$	HL value	(D)-ratio, $c=\frac{1}{8}$	$c=\frac{1}{4}$	$c=\frac{1}{2}$
$5\cdot10^4$	0.397	0.4870	0.4899	–	0.771	0.866
$2\cdot10^5$	0.408	0.4885	0.4856	0.709	0.794	0.879
$6\cdot10^5$	0.415	0.4855	0.4797	0.727	0.803	0.883

The share of V on $\mathfrak{M}$ is 0.0000 for $c=\frac{1}{8}$ and 0.003–0.005 for $c=\frac{1}{4}$ (Theorem 3.4); the mass of $\mathfrak{M}$ agrees with the Hardy–Littlewood value to 0.006 (statement (M) at these sizes); and the ratio in (D), $\int_{\mathfrak{m}}|P|^2|F|^2/(\int_{\mathfrak{m}}|P|^2\int|F|^2)$, is 0.71–0.88 and increases with X for each fixed c, as it must if the deficit is carried by the transition zone $c\sqrt{X}\le q\le C\sqrt{X}$, whose share of the mass of $|P|^2$ is $O(1/\log X)$. Fitting $V/\mathrm{Diag}\cdot(\log X+1.3325)=a\log X+b$ to the three values gives $a=0.504$ and 0.505 on the two intervals.

Remark 3.5 (The dual Titchmarsh input for weight one). We checked how far the proof of [1, Theorem 1.3] uses the trivial nebentypus. The form φ enters only through (i) the Rankin–Selberg bound $\sum_{n\le X}|\lambda(n)|^2\ll(NX)^{\varepsilon}X$, (ii) the Bessel transforms of their Lemma 7.5, and (iii) the Voronoi formula with additive twists, their Lemma 7.4, quoted from the reference [BMN] of their bibliography "in the case of trivial nebentypus"; the Kloosterman sums produced by (iii) are then treated by their Theorem 2.4 through the Kuznetsov formula. For ϑ_ψ, (i) holds, (ii) holds with $J_{k-1}=J_0$ (and the Ramanujan bound holds by Deligne–Serre). For (iii) with the nebentypus $\omega=\chi_{-D}$ and $(q,D)=1$ the computation in the proof of Lemma 3.1 gives the Voronoi formula with the dual twist $e(-n\overline{aD}/q)$ and the additional factor $\omega(q)$, independent of a but not of q, so that the sums of Kloosterman sums in their Section 8 acquire the weight $\omega(c)$: they become Kloosterman sums for the pair of cusps $(\infty,0)$ of $\Gamma_0(D)$ with the character ω, and their Theorem 2.4 has to be re-proved with the Kuznetsov formula for this pair of cusps and nebentypus; for $D\mid q$ the Voronoi coefficients of their Lemma 7.4 depend on a_1 through a character and the Kloosterman sums become twisted. The tools for both exist (the Kuznetsov formula for general cusps and characters, the bound $\theta\le7/64$ for every nebentypus), but the extension is not in the literature as far as we know and we have not carried it out; Hypothesis 1.4 stays a hypothesis.

Data and code

`code/linnik_class.py` (Table 1, outputs `results/linnik_class_23.txt`, `_31.txt`),
`code/linnik_class_variance.py` (mean square, diagonal, the Wilton supremum:

$\sup_\alpha |\sum_{m\le Y} c(m)e(m\alpha)| = 1.51\sqrt{Y}$, $1.56\sqrt{Y}$, $1.58\sqrt{Y}$ at $Y = 10^5, 10^6, 4\cdot 10^6$ for $D = 23$), `code/linnik_class_arcs.py` (the depletion profile $\rho(q)$ and the model), `code/linnik_class_smooth.py` (the smoothed variance against the β-integrated model).

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